

Rwanda Polytechnic, Kigali College

ETT Y4 BTech

Student name: LUKOLOLA NGANDA Bruno

Reg Number: 26RP00283

Machine Learning

House Price Prediction with Multiple Linear Regression

1. Introduction and objective

A real-estate agency in Rwanda wants a quick tool that estimates the market price of a house from a few simple characteristics. The objective of this project is to clean a raw dataset of 125 past sales, train a multiple linear regression model that predicts the price in million RWF from area, bedrooms, bathrooms, house age, distance to the city centre, parking spaces and neighbourhood, evaluate and interpret the model, save it as *house_price_model.sav*, and publish it as a Streamlit web app whose source code is on GitHub.

2. Data cleaning decisions

The raw file was inspected for missing values, duplicates, impossible values and text categories. The decisions are summarised below (full justification is in the notebook). The cleaned dataset has 115 rows.

Problem found	Where	Action taken	Reason
Duplicate records	5 identical rows at the end of the file (IDs 1011, 1026, 1041, 1011, 1026)	Removed	Copies over-weight some houses and leak between train and test sets
Missing target	House_Price_Million_RWF, 1 row (ID 1080)	Row dropped	A target value must not be invented
Missing	Area (2), Bedrooms,	Median	Keeps rows in a tiny dataset;

numeric predictors	Bathrooms, Age, Distance, Parking (1 each)	imputation inside the Pipeline	median is robust to outliers; fitted on train only
Missing category	Neighborhood, 1 row (ID 1064)	Most-frequent imputation inside the Pipeline	Only one row; mode is the simplest safe choice
Impossible area	Area_m2 = 850 and 1200 (IDs 1006, 1019)	Rows removed	Not credible for 3 to 4 bedroom houses; true value unknown
Impossible price	Price = 950 and 1200 (IDs 1073, 1092)	Rows removed	About 5x above comparable houses; target error cannot be corrected
Extreme age	House_Age_Years = 75 and 49.5 (IDs 1034, 1063)	Capped at upper IQR fence	Possible but high-leverage points
Extreme distance	Distance_to_City_k m = 85 (ID 1048, central district)	Capped at upper IQR fence	Not credible for Nyarugenge; keeps other info of the row
Wrong data types	Bedrooms, Bathrooms, Parking stored as float	Left as numeric (whole numbers)	Float only because of NaN; no effect on the model
Text category	Neighborhood (6 levels)	One-hot encoded (drop first)	Linear regression needs numeric inputs; dropping one level avoids the dummy-variable trap
Identifier	House_ID	Excluded from the predictors	It is a label, not a feature

Key charts

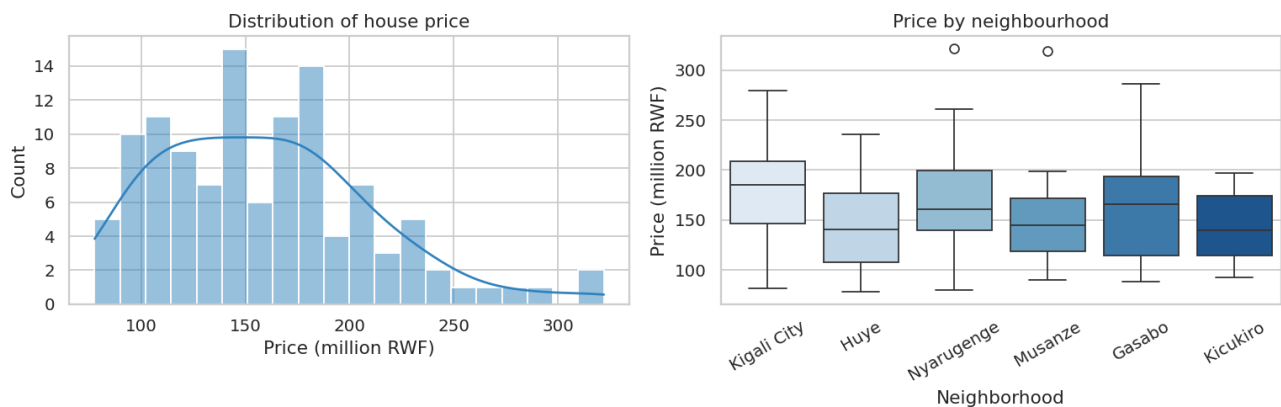
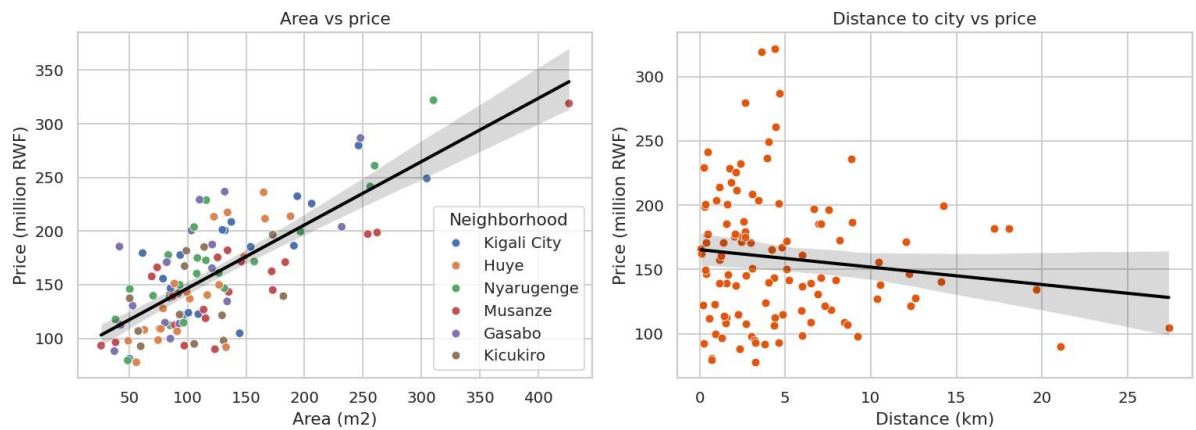


Figure 1. Prices are mildly right-skewed (most between 80 and 220 million RWF). Kigali City has the



highest

median price, but the neighbourhood ranges overlap strongly.

Figure 2. Price rises roughly linearly with floor area (supporting a linear model) and falls slightly with distance from the centre.



Figure 3. Area, bathrooms and bedrooms correlate most strongly with price; bedrooms and bathrooms are very strongly correlated with each other (r about 0.92).

3. Model results

The predictors are all columns except House_ID and the price. A scikit-learn Pipeline combines median imputation for numeric columns, most-frequent imputation and one-hot encoding (first category, Gasabo, dropped) for Neighborhood, and LinearRegression. The data were split 80/20 with random_state=42 (92 training and 23 test houses). The intercept is 31.43 million RWF.

Table 2. Coefficients

Variable	Coefficient
Area_m2	0.549
Bedrooms	6.315
Bathrooms	16.637
House_Age_Years	-0.803
Distance_to_City_km	-0.858
Parking_Spaces	6.008
Neighborhood_Huye	-15.398
Neighborhood_Kicukiro	-14.227
Neighborhood_Kigali City	12.788
Neighborhood_Musanze	-18.617
Neighborhood_Nyarugenge	5.391

Table 3. Evaluation (million RWF)

Set	n	R2	Adj. R2	MAE	RMS E
Trainin g	92	0.87 5	0.858	14.6 6	17.77
Test	23	0.78 4	0.568	18.9 9	23.94

Table 4. Variance inflation factors

Variable	VIF
Area_m2	1.12
Bedrooms	7.73
Bathrooms	7.89
House_Age_Years	1.11
Distance_to_City_km	1.15
Parking_Spaces	1.17
Neighborhood_Huye	2.00
Neighborhood_Kicukiro	1.71
Neighborhood_Kigali City	1.75
Neighborhood_Musanze	1.87
Neighborhood_Nyarugenge	1.77

The model explains about 88% of the price variation in the training set and about 78% in the test set (5-fold cross-validated R^2 is about 0.79); the typical test error (RMSE) is about 24 million RWF. Bedrooms and Bathrooms have VIF values of about 7.7 and 7.9 because they are strongly related; all other VIFs are below 2.1.

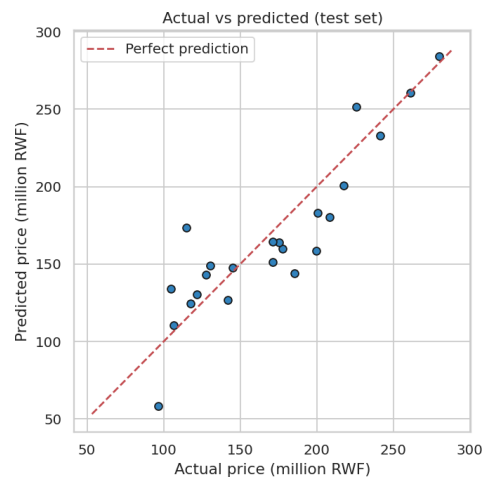


Figure 4. Actual vs predicted prices on the test set: points follow the diagonal reasonably well; a few houses are missed by 40 to 60 million RWF.

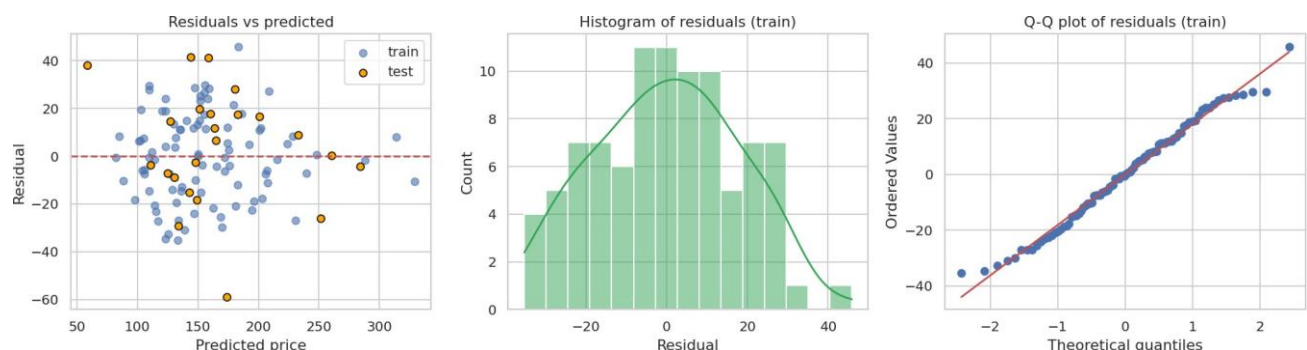


Figure 5. Residual diagnostics: no funnel shape, a roughly bell-shaped histogram and a Q-Q plot close to the line. Shapiro-Wilk $p = 0.40$, Breusch-Pagan $p = 0.71$, Durbin-Watson = 2.10.

Interpretation of coefficients

- Area (+0.55): holding everything else constant, each extra square metre adds about 0.55 million RWF; 10 extra m² add about 5.5 million RWF.
- House age (-0.80): each extra year of age lowers the price by about 0.80 million RWF.
- Distance to the centre (-0.86): each extra kilometre lowers the price by about 0.86 million RWF.
- Parking (+6.0): each extra parking space adds about 6.0 million RWF.
- Neighbourhood (reference Gasabo): an otherwise identical house is about 12.8 million RWF dearer in Kigali City and about 18.6 million RWF cheaper in Musanze.

- Bathrooms (+16.6) and Bedrooms (+6.3): positive, but because the two are strongly correlated their individual sizes are unreliable; only their combined effect is trustworthy.

Assumption checks

Linearity: reasonably satisfied (straight-line Area-Price pattern, no curve in the residuals).

Independence: reasonably satisfied (separate houses, Durbin-Watson about 2.1).

Constant variance: reasonably satisfied (no funnel, Breusch-Pagan $p = 0.71$). Normal residuals: reasonably satisfied (Q-Q plot, Shapiro-Wilk $p = 0.40$).

Multicollinearity: no severe problem except the Bedrooms/Bathrooms pair (VIF about 7.8).

4. The Streamlit app

app.py loads the saved pipeline with `@st.cache_resource` and a relative path, offers input widgets for all seven features with ranges taken from the cleaned data, and a Predict button that builds a one-row DataFrame with the training column names, calls `model.predict()` and shows the price in million RWF. It warns the user when a value is outside the training range. Example: a 120 m² house with 3 bedrooms, 2 bathrooms, 5 years old, 4 km from the centre, 1 parking space in Kicukiro is predicted at about 133.9 million RWF.

5. Conclusion, limitations and ideas for improvement

Conclusion. After careful cleaning, a multiple linear regression pipeline explains most of the variation in house prices (test $R^2 = 0.78$, RMSE = 24 million RWF, MAE = 19 million RWF). Area, bathrooms, parking and neighbourhood raise the price, while age and distance from the centre lower it. The model is saved as a single .sav pipeline that accepts raw inputs and is served through a Streamlit app.

Limitations. The dataset is very small (115 clean houses, 23 in the test set), so metrics are uncertain and extrapolation outside the training ranges (above about 430 m², older than 35 years, further than 27 km) is unreliable.

The model is purely linear and omits factors such as finishing quality, plot size and sale date; it also cannot capture interactions such as a different price per m² in different neighbourhoods. Bedrooms and Bathrooms are collinear, so their individual coefficients are unstable. Four records had to be removed as impossible values.

Ideas for improvement. Collect more sales; merge Bedrooms and Bathrooms into one variable or keep only one; add interaction terms (Area x Neighborhood) or try regularised (Ridge/Lasso) and tree-based models; model the log of the price; use cross-validation for model selection; add prediction intervals to the app and retrain regularly with new sales.